UNSUPERVISED LEARNING - CLUSTERING ALGORITHMS

- **1. K-Means**

  Working Principle:

- Choose K clusters (you decide number).
- Place K points randomly (centroids).
- Assign each data point to nearest centroid.
- Update centroid as average of points.
- Repeat until stable.

  Advantages:
- Simple, fast
- Works well for spherical (round) clusters

  Disadvantages:

- Must know K beforehand

- Bad for irregular shapes or noise

  Example:
- Imagine you have students’ marks in Math & English. You want 2 groups: "Good in both" and "Weak in both". K-means will separate them into 2 round clusters.

- **2. Affinity Propagation**

  Working Principle:

- Unlike K-means, you don’t choose K.

- Each point sends "messages" to others:

- "Responsibility" = how suitable you are as my cluster center

- "Availability" = how available you are to be my center

- Points that get most "votes" become cluster centers (exemplars).

Advantages:
- No need to specify K
- Finds real data points as cluster centers

Disadvantages:
- Slower than K-means
- Memory heavy for big data

Example:
- Friends deciding a meeting place. Everyone votes for who could be the meeting leader, and the one with most mutual support becomes the leader (cluster center).

- **3. Mean Shift**

Working Principle:

- Place a window (circle) around each point.

- Slide (shift) the window to the area of highest data density (mode).

- Points shifting to the same mode = same cluster.

Advantages:

- Automatically finds number of clusters

- Good for irregular shapes

Disadvantages:

- Slow on large data
- Bandwidth (window size) tricky to choose

Example:

- Think of ants gathering on sugar grains. Each ant moves to where the most sugar is → groups form around sugar piles.

🔷 4. Spectral Clustering

Working Principle:

- Build a similarity graph (points as nodes, edges = similarity).
- Cut the graph into clusters using eigenvalues (linear algebra).

Advantages:

- Works for complex, non-spherical shapes
- Good when clusters are connected in graph form

Disadvantages:

- Computationally heavy
- Needs similarity measure

Example:

- Imagine a social network. People connected strongly (friends) get grouped into communities.

🔷 5. Hierarchical Clustering

- **Working Principle:**

  Two types:

- 1)**Agglomerative**: Start with each point → merge closest pairs → repeat
- 2)**Divisive**: Start with one big group → split into smaller groups
- Creates a tree (dendrogram) of clusters.

  Advantages:

- No need to choose K initially
- Tree structure helps analyze relationships

  Disadvantages:

- Once merged/split, can’t undo
- Slow for very large data

  Example:

- Family tree: cousins grouped → families → clans → tribe.

◈ 6. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

Working Principle:

- Groups together points with many nearby neighbors (dense regions).
- Points far away = noise (outliers).

- Needs 2 parameters:
- eps (radius)
- minPts (minimum points in neighborhood)

Advantages:

- Finds clusters of any shape
- Handles noise well

Disadvantages:

- Choosing eps & minPts tricky
- Not good with varying densities

Example:

- Stars in the sky: Constellations form where stars are dense. Lonely stars = outliers.

◈ 7**. HDBSCAN (Hierarchical DBSCAN)**

Working Principle:

- Extension of DBSCAN
- Instead of fixed eps, it builds hierarchy of densities → chooses stable clusters.

Advantages:

- Works with varying density
- Less sensitive to parameter tuning

Disadvantages:

- More complex to understand than DBSCAN
- Slower

Example:

- In a city map, some areas have dense shops, some medium, some sparse. HDBSCAN can find all meaningful clusters, unlike DBSCAN.

◈ 8. **OPTICS (Ordering Points To Identify Clustering Structure)**

Working Principle:

- Similar to DBSCAN but instead of fixed eps, it orders points by "reachability distance".
- You can extract clusters of different densities.

Advantages:

- Handles varying density better than DBSCAN
- Provides clustering structure at multiple levels

Disadvantages:

- More complex output (reachability plot needed)
- Slower

Example:

- Like topographic map: valleys = clusters, mountains = gaps.

◈ 9. BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies)

Working Principle:

- Incrementally builds a tree (CF tree) to summarize data.
- Then clusters the summaries instead of all points.

Advantages:

- Very fast, memory-efficient for big data
- Works well as preprocessing

Disadvantages:

- Works best for spherical clusters
- Approximation may lose details

Example:

- Instead of counting every grain in a rice sack, you group handfuls → then cluster handfuls.

Quick Mnemonics to remember:

K-Means → Fixed K, round groups

Affinity Propagation → Voting for leaders

Mean Shift → Move to densest sugar pile

Spectral → Social network communities

Hierarchical → Family tree

DBSCAN → Stars & constellations

HDBSCAN → DBSCAN with flexible density

OPTICS → Topographic map valleys

BIRCH → Rice sack summarization